■ MDB Agent System

Installation & User Guide

A complete step-by-step guide for installing Python, setting up the Flask application, and using all system features.

Platform	Windows
Language	Html, CSS, Python
Framework	Flask (Web Application)
Database	SQLite (built-in)

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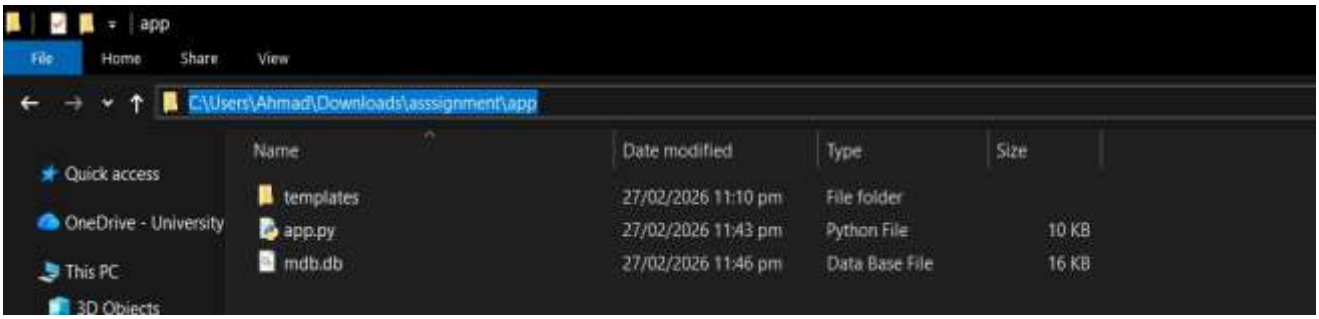
1. Installing Python

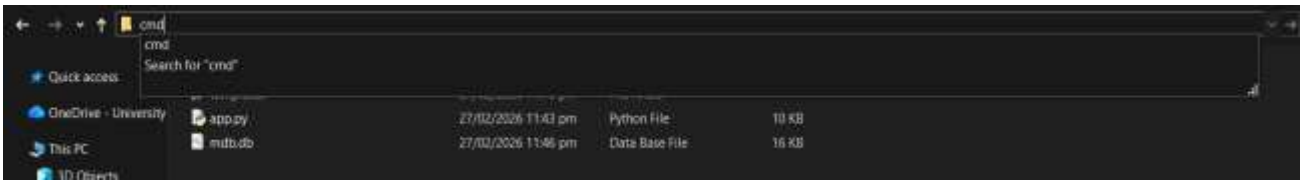
Windows

1. Open your browser and Download Python installer.
   https://www.python.org/ftp/python/3.14.3/python-3.14.3-amd64.exe
2. Run the downloaded installer (.exe file).
3. **Important:** On the first screen, check the box that says **"Add Python to PATH"** before clicking Install Now.
4. Click **Install Now** and wait for the installation to complete.
5. Click **Close** when done.

2. Installing Dependencies

Just run install flask file inside app directory it will auto install it.





3. Running the Application

After installing python and Flask just run start app file.

You should see output like:

```
* Running on
  http://127.0.0.1:5000 * Debug
  mode: on
* Restarting with stat
```

Open your web browser and go to: **http://127.0.0.1:5000** or **http://localhost:5000**

To stop the server, press **Ctrl + C** in the terminal or just close it.

First run note: On the first run, the application automatically creates the SQLite database file (**mdb.db**) and inserts a default administrator account. You do not need to set up the database manually.

4. First-Time Setup & Default Admin Account

A default administrator account is automatically created when the app runs for the first time:

Field	Value
Email	maryam@admin.com
Password	admin123
Role	Admin

5. Using the Application

5.1 Registration (Students)

New students can create their own accounts by navigating to the registration page.
Instructor and Admin accounts can only be created by an existing Admin.

To register as a student:

1. Go to **http://localhost:5000/register** or click **"Register as Student"** on the login page.
2. Fill in your **Full Name**, **Email Address**, **Password**, and **Confirm Password**.
3. Password must be **at least 6 characters** long.
4. Both password fields must match.
5. Click **Create Account**.
6. You will be redirected to the login page with a success message.

Note: Each email address can only be registered once. If you see an error saying "Email already registered", the address is already in use.

5.2 Logging In

All users (students, instructors, and admins) log in from the same login page:

1. Navigate to **http://localhost:5000** — you will be automatically redirected to login.
2. Enter your registered **Email Address** and **Password**.
3. Click **Sign In**.
4. You will be automatically redirected to the dashboard appropriate for your role.

If your account has been blocked by an administrator, you will see a message: "Your account has been blocked. Contact administrator." and will not be able to log in.

5.3 Student Dashboard

- Student Dashboard is an AI-based learning interface after login
- Students can ask academic questions using the **AI Query box**
- Queries are sent to the backend Flask system for processing
- System checks if the question is academic or not

- AI searches uploaded PDF handouts for relevant answers
- If match is found → answer is generated from handouts
- If no match → “No Answer Found” message is shown
- “My MDB History” stores all previous queries and responses
- Each record shows:
  - Query text
  - Status (Answered / No Answer Found)
  - AI response
- Right sidebar shows:
  - Student profile info
  - Logout button
- Dashboard helps students learn using AI instead of manual searching

5.4 Instructor Dashboard

- Instructor uploads **PDF handouts**
- System stores uploaded files in the database
- Text is automatically extracted from PDFs
- Extracted text becomes the **AI knowledge base**
- Handouts must be clear and text-based for proper processing
- AI uses these handouts to answer student questions
- If no matching content is found, AI cannot generate a direct answer
- Instructor is responsible for updating and maintaining handouts

5.5 Admin Panel

Admins are redirected to **/admin/panel** after login. This is the most powerful section of the application. The admin panel is divided into three main sections:

Overview Stats

The top of the admin panel shows four statistics cards:

- **Total Users** (blue) – total number of accounts in the system.
- **Students** (green) – count of accounts with student role.
- **Instructors** (orange) – count of accounts with instructor role.
- **Blocked** (red) – count of blocked accounts.

Create New User

Admins can create accounts for any role. Fill in the form with:

- **Full Name** – the user's display name.
- **Email Address** – must be unique in the system.
- **Password** – the initial password for the new user.
- **Role** – select Student, Instructor, or Admin from the dropdown.

Click **Create User** to add the account. The new user will appear in the table below.

All Users Table

The users table lists every account in the system with the following columns:

Column	Description
#	The unique numeric ID of the user
Name	The user's full name
Email	The user's email address (login identifier)
Role	Dropdown to change role — save with the Set button
Status	Active (green) or Blocked (red)
Registered	Date the account was created
Actions	Block/Unblock button and Delete button